Week 12: 16– 20 November 2026

Subject: Mathematics **Unit Title:** Linear Functions and Equations **Grade :** 8 Pre AP Algebra 1

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative/تكويني Summative /ختامي Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: 1.7 Sequences, Functions, and Graphs **Lesson 2 1.8 Interpreting Different Forms of a Linear Equation**	• 1.1.2 Construct and use arithmetic sequences to represent linear relationships. • 1.1.5 Construct and apply linear functions to solve problems given mathematical or applied contexts. • 1.2.1 Identify and interpret key characteristics of linear functions and their graphs. • 1.2.2 Generate an algebraic representation of a linear function given a verbal description, graphical representation, or tabular representation. • 1.2.3 Translate between different algebraic forms of a linear equation.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.1 Make goals clear, visible, and supported by short-term objectives.	Reflection prompts, notebooks The resources will be available on student's Teams channel
Lesson 3: 1.9 Writing Linear Equations in Standard Form	• I can **model** relationships between two variables by **identifying** constraints in a real-world situation and **writing** a linear equation in $standard\ form$: $(\boldsymbol{Ax} + \boldsymbol{By} = \boldsymbol{C})$.		
Lesson 4 1.9 Representing and Interpreting Solutions of Linear Equations	• I can **analyze** solutions to a linear equation by **generating** solution pairs, **graphing** them on a coordinate plane, and **interpreting** the slope and intercepts in context		

Teacher's notes/ Reminders ملاحظات المعلم والتذكير	**Formatives take place during lesson time, ALWAYS be prepared.** Bring your stationary: Penic/Pen and Calculator and be on time. **Remember** we will be using our notebooks. ***Devices will only be used on Friday's, unless otherwise communicated by your teacher.*** Communication takes place via **TEAMS** or **Toddle**. We will be introducing our AP Platform soon, **College Board** **TIMMS Preparation: Students have a school link to enable them to practice for the upcoming TIMMS testing in 2027. Remember this is an international external test that occurs every few years, testing only grade 8's.** **Homework** is assigned through TEAMS/TODDLE as practice from teachers. **Performance (Grasps Tasks) are due 23 November 2026**

Technology Integration دمج التكنولوجيا	Resources shared on Microsoft Teams, College Board
Cross-curricular links الروابط بين المواد الدراسية	**Science**: Measurement and unit conversions. **Physics**: Speed, rate, and proportional relationships. **Geography**: Metric and customary measurement systems. **Engineering**/Design: Using mathematical models to solve real-world problems.
External Assessment Integration دمج التقييمات الخارجية	All content is covered in AP Past Exams/ College Board Assessments
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Wellbeing:** • Confidence in solving real-world maths problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussions. • Self-awareness by setting and reflecting on learning goals. **IB ATL Skills:** • **Thinking:** Analyze patterns and relationships. • **Communication:** Explain and justify mathematical reasoning. • **Research:** Collect, organize, and interpret data. • **Social:** Collaborate and discuss problem-solving strategies
Standards المعايير	• **A.CED.2** • **F.IF.3, 5** • **F.BF.1, 2** • **F.LE.2**
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can different representations of a linear relationship help us understand and solve real-world problems?

Success Criteria معايير النجاح (Measurable)	Lesson 1 **All Students Must:** connect arithmetic sequences to linear functions and represent them graphically. **Most Students Should:** determine the rate of change and initial value from tables, graphs, and sequences. **Some Students Could:** construct and apply linear functions to solve real-world problems involving constant rates of change. Lesson 2: **All Students Must:** identify slope and y-intercepts from graphs, tables, and equations. **Most Students Should:** write linear equations in slope-intercept form and interpret the meaning of the slope and intercept in context. **Some Students Could:** generate linear equations from graphical, tabular, and verbal representations. Lesson 3: **All Students Must:** Identify the variables, coefficients, and constraint in a real-world scenario and write a correct equation in standard form.. **Most Students Should:** Explain what each value and variable represents in the context and justify why the equation models the situation. **Some Students Could:** Create their own real-world scenario, develop a standard-form equation, and defend how the equation represents the constraints of the situation. Lesson 4: **All Students Must:** Generate multiple solution pairs for a linear equation and plot them accurately on a graph. **Most Students Should:** Determine and explain the meaning of the slope and intercepts using the context of the problem. **Some Students Could** Evaluate whether points on the graph are meaningful solutions in the context, justify their reasoning, and compare different representations (equation, table, and graph).
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Respect الاحترام Heritage التراث